Ethan Ragbir

Undergraduate Researcher in Physics, The College of New Jersey

ragbire1@tcnj.edu · (718) 607-6955 · New Jersey, USA

Updated November 2026

Education

2024–2028 **B.S. in Physics**, The College of New Jersey (TCNJ).
Expected May 2028. Academic adviser: Dr. Tuan Nguyen.
Selected coursework: General Relativity; Quantum Mechanics; Electrodynamics; Electronics; Statistical Mechanics; Partial Differential Equations; Physics of Cloud and Climate.

Research interests

Experimental nuclear and particle physics; atmospheric and cloud microphysics; scientific instrumentation and detector development; computational data analysis and machine learning.

Research experience

Jan 2026–present **Undergraduate Researcher**, The College of New Jersey.
Adviser: Dr. Salina Ali.
Analyze Jefferson Lab Neutral Particle Spectrometer data to reconstruct neutral-pion ($\pi^0 \to \gamma\gamma$) events; develop C++ and CERN ROOT workflows for event selection, detector studies, and simulations. Collaborate with researchers at Jefferson Lab and the Catholic University of America, using high-performance computing for large datasets.

May–Aug 2026 **Undergraduate Intern**, Thomas Jefferson National Accelerator Facility (Jefferson Lab).
Supported experimental nuclear-physics research and detector/data-acquisition work associated with high-energy electron-beam experiments. Participated in quality-assurance measurements of lead-tungstate calorimeter crystals, including optical-transmission studies.

May–July 2026 **Research Experiences for Undergraduates (REU) Researcher**, The Catholic University of America.
Adviser: Dr. Tanja Horn.
Investigated neutral-pion electroproduction with liquid-deuterium data from Jefferson Lab Hall C, applying numerical analysis and two-photon invariant-mass reconstruction to study candidate events and experimental backgrounds.

Apr 2025–present **Undergraduate Researcher**, The College of New Jersey.
Adviser: Dr. Nate Magee.
Apply cryogenic scanning electron microscopy (cryo-SEM) to characterize laboratory-grown ice-crystal morphology, contributing experimental observations relevant to cirrus-cloud microphysics and climate research.

June 2025–Jan 2026 **Research Collaborator**, Rutgers University, High Energy Experimental Physics (HEX) Group.
Adviser: Sunil Somalwar.
Developed a neural-network analysis pipeline using Monte Carlo simulations for CMS supersymmetry searches; studied electroweak and R-parity-violating decay signatures using high-energy physics datasets.

Professional and experimental experience

Feb 2026–present **Physics Laboratory Technician**, The College of New Jersey.
Prepare and maintain experiments for introductory mechanics, electricity and magnetism, and modern physics laboratories. Troubleshoot multimeters, power supplies, motion apparatus, and circuits; perform basic soldering and support faculty during lab sessions.

Apr–Sept 2025 **Research Intern, Machine Learning**, General Dynamics.
Optimized jet-propulsion dynamic models by combining retrieval-augmented generation (RAG) with physics-based modeling approaches.

Apr–Sept 2024 **Systems Engineering Intern**, Lockheed Martin Advanced Technology Laboratories.
Integrated simultaneous localization and mapping (SLAM) algorithms and sensor-fusion pipelines for autonomous aerospace-navigation systems.

Selected conference talks and presentations

Oct 2026 **Neutron Structure through Electroproduction.**
Talk, The College of New Jersey Colloqium. E.Ragbir, T.Horn, S.Ali

Oct 2026 **Probing Neutron Structure through Electroproduction.**
Poster presentation, American Physical Society Division of Nuclear Physics meeting.

Aug 2026 **Cryo-SEM Imaging of Lab-Grown Cirrus Analog Crystals.**
E. Ragbir, E. Jansen, and G. Lampmann. Poster presentation, American Meteorological Society.

July 2026 **Probing Neutron Structure through Electroproduction.**
E. Ragbir, T. Horn, A. Singh, and J. Crafts. Talk and poster, REU research presentations, Catholic University of America and Jefferson Lab.

Apr 2026 **Cryo-SEM Imaging of Lab-Grown Cirrus Analog Crystals.**
E. Ragbir, E. Jansen, and G. Lampmann. Poster, Celebration of Student Achievement, The College of New Jersey.

Dec 2025 **Cloud in a Box.**
E. Ragbir, E. Jansen, and G. Lampmann. Departmental research presentation, The College of New Jersey.

Technical skills

Programming Python, C++, C, R, LaTeX.

Scientific software CERN ROOT, ImageJ, Git/GitHub, Visual Studio Code, Microsoft Excel; simulation and scientific computing workflows.

Research methods Experimental data analysis, detector-data processing, Monte Carlo methods, statistical analysis, data visualization, scanning electron microscopy, electronics troubleshooting.

Leadership and activities

Aug 2024–present **Bonner Community Scholar and Site Leader**, The College of New Jersey.
Contribute more than 300 hours of community service annually across the Trenton and Ewing communities. Coordinate a team of approximately eight volunteers, organizing assignments, outreach, and daily service operations.

Fall 2024–present **Vice President**, Physics Club Executive Board, The College of New Jersey.
Coordinate outreach and departmental events, including public astronomy events and physics demonstrations.

Fall 2024–present **Competitor**, William Lowell Putnam Mathematical Competition Team.
Prepare for collegiate mathematics competition, with emphasis on advanced mathematical reasoning, combinatorics, and analysis.